A Case Study Report

on

"TITANIC SURVIVAL PREDICTION USING LOGISTIC REGRESSION"

for the course

Fundamentals Of Data Science (22CSC59)

Submitted in partial fulfillment of the requirements for the award of degree

BACHELOR OF ENGINEERING

in

COMPUTER SCIENCE AND ENGINEERING

by

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1. PROBLEM STATEMENT

The Titanic Survival Prediction case study focuses on predicting whether a passenger survived the Titanic disaster using machine learning techniques. The prediction is based on passenger details such as age, gender, passenger class, ticket fare, and family information.

The main objective is to develop a Logistic Regression model that learns patterns from historical passenger data and predicts survival outcomes for new passenger details. The project involves analyzing the dataset, handling missing values, encoding categorical variables, training the model, and evaluating its performance using appropriate classification metrics.

The trained model is also integrated into a Streamlit web application, allowing users to enter passenger details and obtain a survival prediction through a simple interface.

2. METHODOLOGY

The methodology followed in this project consists of several steps, beginning with data collection and ending with the deployment of the trained model.

Step 1: Dataset Loading

The Titanic dataset is loaded into a Pandas DataFrame using Python. The dataset is examined to understand its structure, features, data types, and missing values.

```
df = pd.read_csv("titanic (1).csv")

print("First 5 rows:")
display(df.head())

print("Dataset shape:", df.shape)
```

```
First 5 rows:
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

```
Dataset shape: (891, 12)
```

Figure 1. Loading and Inspecting the Titanic Dataset

Step 2: Exploratory Data Analysis

Exploratory Data Analysis (EDA) is performed to understand the distribution of passenger information and identify patterns related to survival. Different visualizations are created to examine survival counts, passenger gender, passenger class, age, and ticket fare.

Step 3: Data Preprocessing

```
# Process numerical data
numeric_transformer = Pipeline(steps=[
    ("imputer", SimpleImputer(strategy="median")),
    ("scaler", StandardScaler())
])

# Process categorical data
categorical_transformer = Pipeline(steps=[
    ("imputer", SimpleImputer(strategy="most_frequent")),
    ("encoder", OneHotEncoder(handle_unknown="ignore"))
])

# Combine preprocessing steps
preprocessor = ColumnTransformer(transformers=[
    ("num", numeric_transformer, numeric_features),
    ("cat", categorical_transformer, categorical_features)
])

print("Preprocessing setup completed!")
```

```
Preprocessing setup completed!
```

Figure 2: Data Preprocessing and Logistic Regression Model Pipeline

Missing numerical values are handled using median imputation, while missing categorical values are filled using the most frequent value. Categorical variables are converted into numerical representations using One-Hot Encoding. Numerical features are scaled using StandardScaler.

Step 4: Model Building

A Logistic Regression model is developed using the Scikit-learn library. The selected features are used as inputs, and the Survived column is used as the target variable. The data is divided into training and testing sets to train and evaluate the model.

```
from sklearn.model_selection import train_test_split
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import OneHotEncoder, StandardScaler
from sklearn.impute import SimpleImputer
from sklearn.linear_model import LogisticRegression

# Select useful features
X = df[["Pclass", "Sex", "Age", "SibSp", "Parch", "Fare", "Embarked"]]
y = df["Survived"]

# Identify numerical and categorical columns
numeric_features = ["Pclass", "Age", "SibSp", "Parch", "Fare"]
categorical_features = ["Sex", "Embarked"]

# Split data into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

print("Training data:", X_train.shape)
print("Testing data:", X_test.shape)
```

```
Training data: (712, 7)
Testing data: (179, 7)
```

Figure 3: Data Preprocessing and Logistic Regression Model Pipeline

Step 5: Model Evaluation

The trained model is evaluated using accuracy, precision, recall, F1-score, and ROC-AUC. An ROC curve is plotted to visualize the model's classification performance. The model coefficients are also examined to understand the relationships between the input features and survival predictions.

Step 6: Deployment

The trained model and preprocessing pipeline are saved using Joblib. A Streamlit application is developed to accept passenger details and display survival predictions. The application is deployed online using Streamlit Community Cloud, with the project files maintained in GitHub.

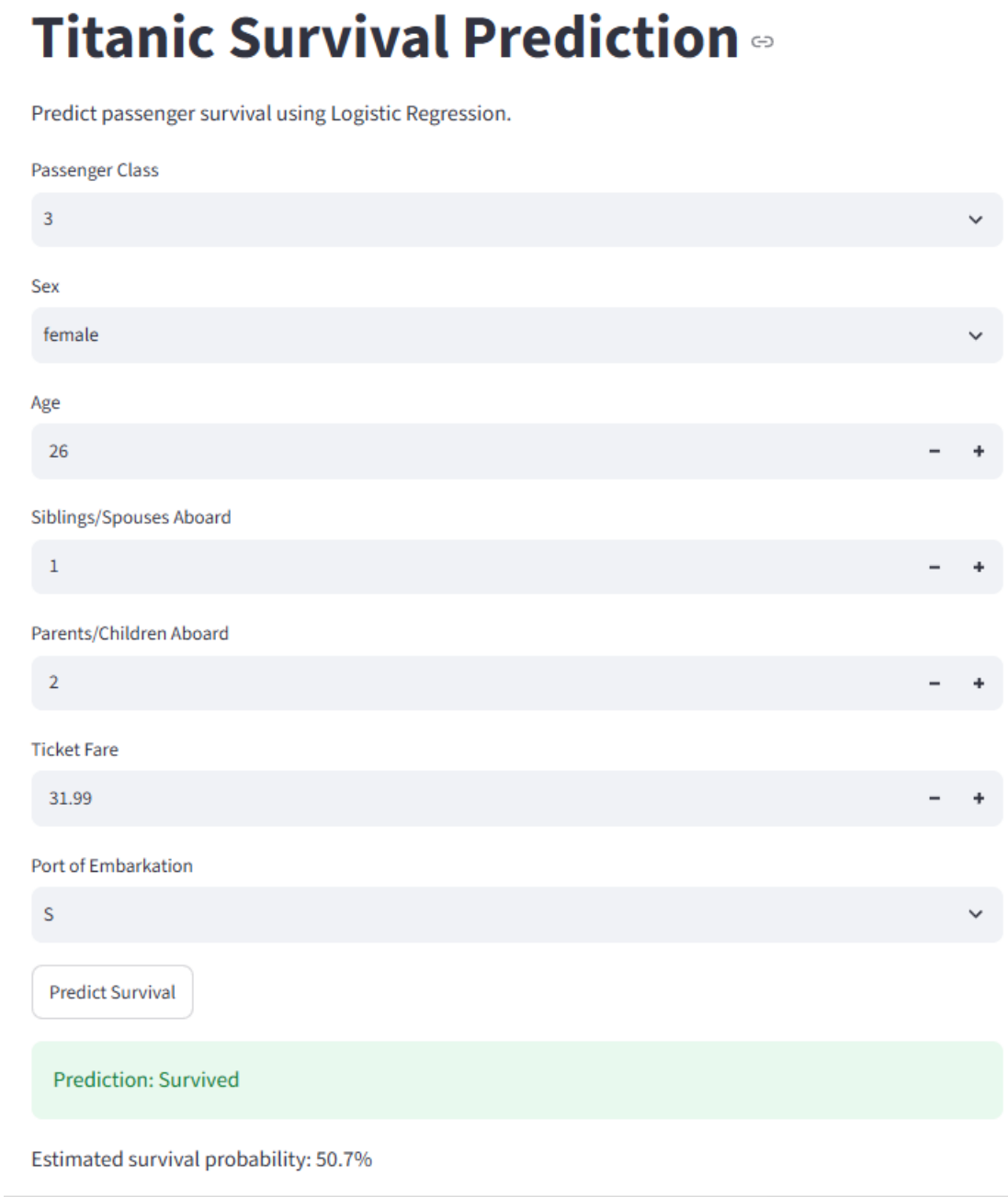


Figure 4: Streamlit Web Application for Titanic Survival Prediction

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3. STATISTICAL FINDINGS

Statistical findings are obtained by analyzing the Titanic dataset and examining the relationships between passenger characteristics and survival. Exploratory Data Analysis (EDA) is used to understand the distribution of passenger data and identify patterns related to survival. Visualizations such as bar plots, histograms, and correlation heatmaps help present these findings clearly.

3.1 Survival Count

```
plt.figure(figsize=(6, 4))
sns.countplot(data=df, x="Survived")
plt.title("Survival Count")
plt.xlabel("Survived (0 = No, 1 = Yes)")
plt.ylabel("Number of Passengers")
plt.show()
```

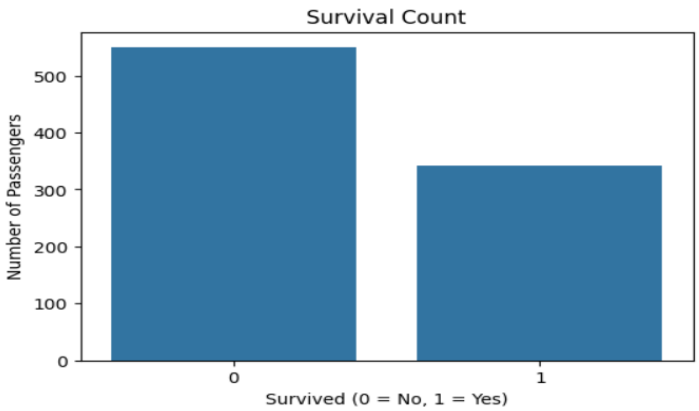


Figure 5. Survival Distribution of Titanic Passengers

Interpretation: This graph shows the number of passengers who survived and those who did not survive the Titanic disaster. It helps us understand the overall survival distribution in the dataset and provides a starting point for analyzing the factors that influenced survival.

3.2 Survival by Gender

```
plt.figure(figsize=(6, 4))
sns.countplot(data=df, x="Sex", hue="Survived")
plt.title("Survival by Gender")
plt.show()
```

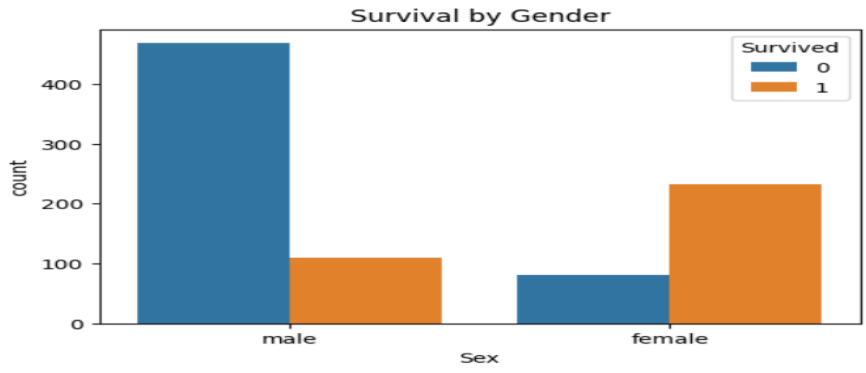


Figure 6. Survival Distribution by Gender

Interpretation: This graph compares the survival outcomes of male and female passengers aboard the Titanic. It helps identify the relationship between gender and survival and shows whether survival outcomes differed between the two groups.

3.3 Survival by Passenger Class

```
plt.figure(figsize=(6, 4))
sns.countplot(data=df, x="Pclass", hue="Survived")
plt.title("Survival by Passenger Class")
plt.show()
```

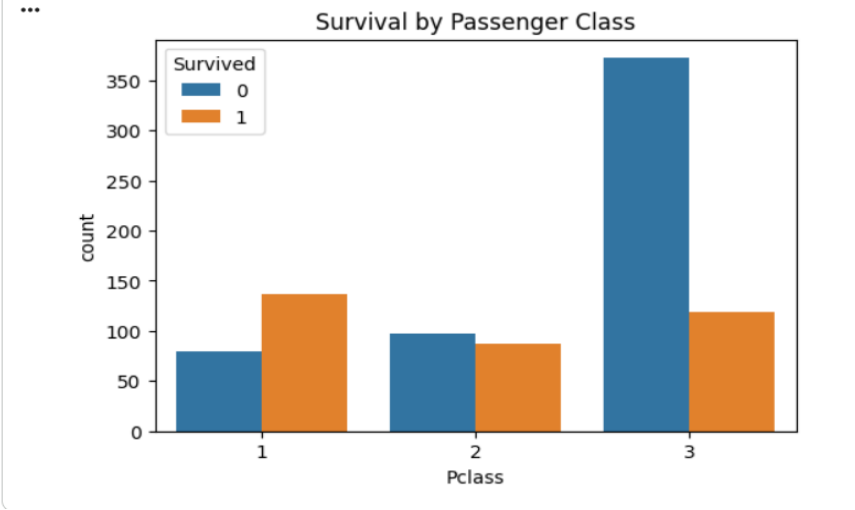


Figure 7. Survival Distribution by Passenger Class

Interpretation: This graph compares the survival outcomes of passengers belonging to different passenger classes (1st, 2nd, and 3rd class). It helps us understand how passenger class was associated with survival during the Titanic disaster. The graph can be used to compare survival patterns across the three classes.

3.4 Age Distribution

```
plt.figure(figsize=(6, 4))
sns.histplot(data=df, x="Age", bins=20, kde=True)
plt.title("Distribution of Passenger Ages")
plt.show()
```

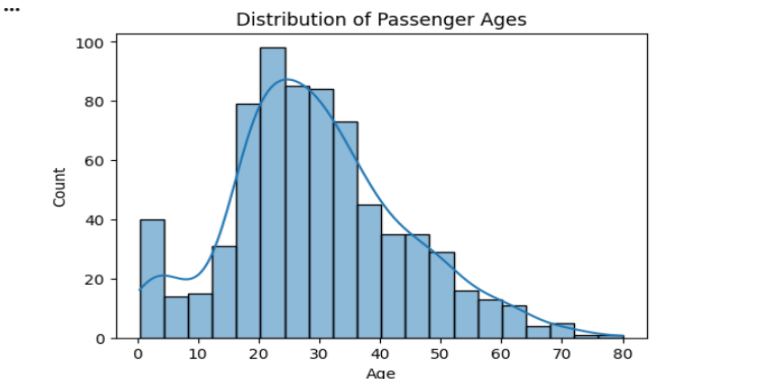


Figure 8. Age Distribution of Titanic Passengers

Interpretation: This histogram shows the distribution of ages among passengers aboard the Titanic. It helps us understand the age groups represented in the dataset and observe which age ranges contain more passengers. This analysis provides a better understanding of the passenger demographics.

3.5 Ticket Fare Distribution

```
plt.figure(figsize=(6, 4))
sns.histplot(data=df, x="Fare", bins=20, kde=True)
plt.title("Distribution of Passenger Ticket Fares")
plt.show()
```

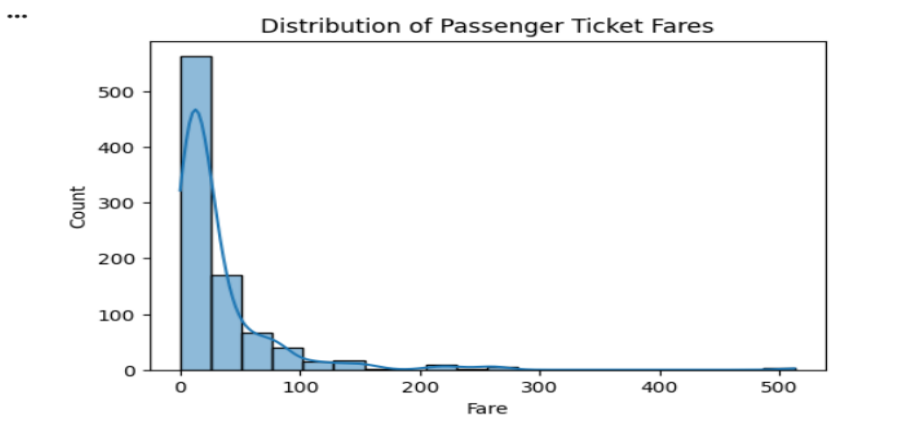


Figure 9. Distribution of Ticket Fares Among Titanic Passengers

Interpretation: This histogram shows the distribution of ticket fares paid by Titanic passengers. It helps identify the range of ticket prices and understand how frequently different fare values occur in the dataset. The graph also helps observe whether the fare distribution is concentrated within a particular range or contains a few unusually high values.

3.6 Correlation Heatmap

```
# Correlation heatmap for numerical features
plt.figure(figsize=(8, 6))
correlation = df[
    ["Survived", "Pclass", "Age", "SibSp", "Parch", "Fare"]
].corr()

sns.heatmap(correlation, annot=True, cmap="coolwarm", fmt=".2f")
plt.title("Correlation Heatmap of Numerical Features")
plt.show()
```

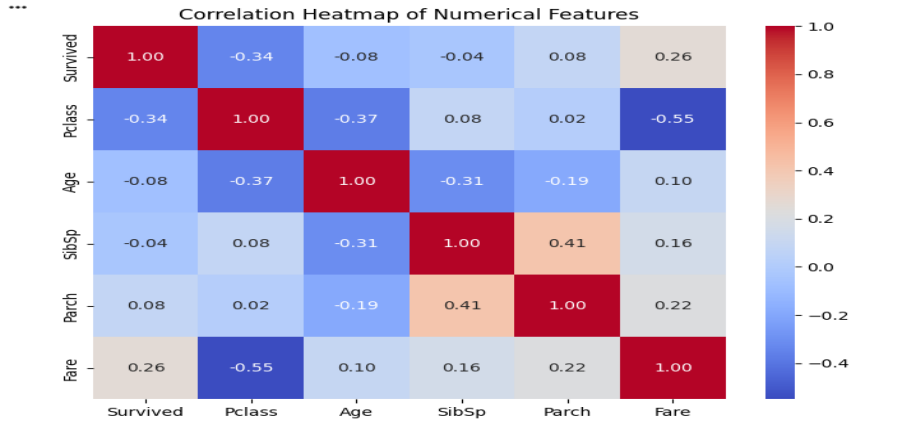


Figure 10. Correlation Heatmap of Numerical Features

Interpretation: The correlation heatmap represents the relationships between numerical features in the Titanic dataset using different colours. It helps identify positive and negative correlations between variables and shows how strongly numerical features are related to one another. This analysis provides useful insights into the dataset before interpreting the model results.

3.7 Box Plot

```
import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(8, 5))
sns.boxplot(x=df["Age"].dropna())

plt.title("Box Plot of Passenger Ages")
plt.xlabel("Age (years)")
plt.grid(axis="x", linestyle="--", alpha=0.5)
plt.show()
```

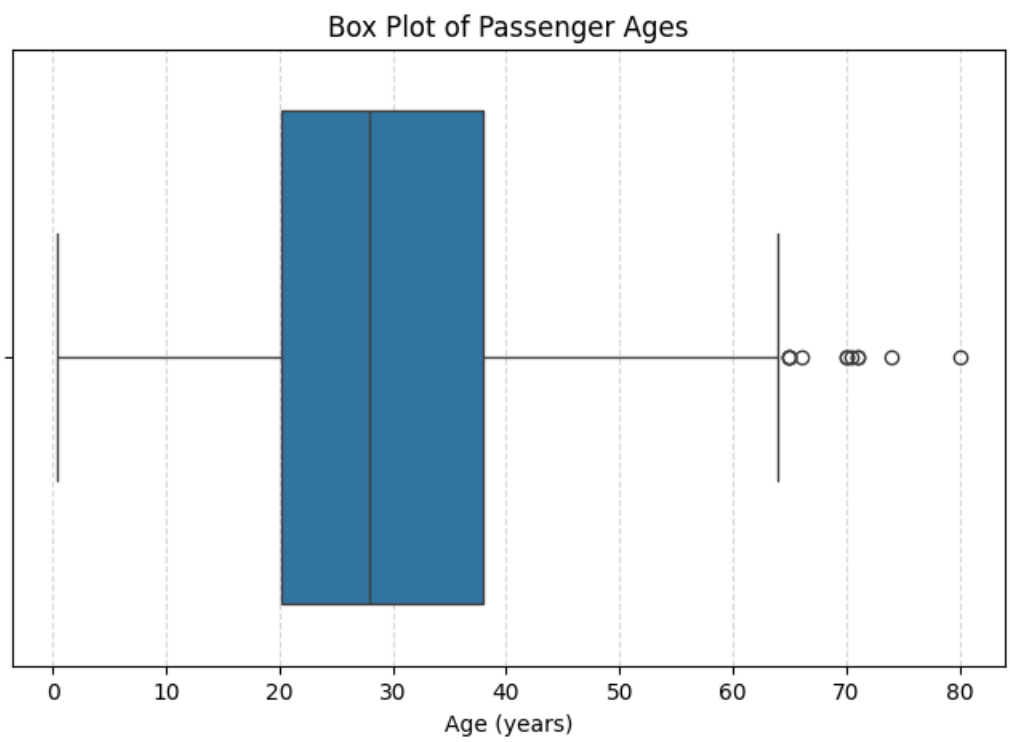


Figure 11. Box Plot of Passenger Ages

Interpretation: The box plot shows the median age and the spread of passenger ages in the Titanic dataset. The points beyond the whiskers represent potential outliers, indicating passengers whose ages differ from the main distribution

4. MODEL EVALUATION

The Logistic Regression model was evaluated using the testing dataset to measure its ability to predict passenger survival. The dataset was divided into training and testing sets, with 80% of the data used for training and 20% for testing. A total of 712 records were used for training, while 179 records were used for testing.

The model's performance was measured using accuracy, precision, recall, F1-score, and ROC-AUC. These metrics help evaluate the correctness of predictions and the model's ability to distinguish between passengers who survived and those who did not.

4.1 Performance Metrics

The Logistic Regression model was evaluated using five classification metrics. The results obtained from the testing dataset are presented below.

Evaluation Metric	Score
Accuracy	80.4%
Precision	79.3%
Recall	66.7%
F1-Score	72.4%
ROC-AUC	84.4%

Interpretation: The model achieved an accuracy of 80.4%, indicating that it correctly classified approximately 80% of the test records. The precision of 79.3% shows that its positive survival predictions were reasonably reliable. The recall of 66.7% indicates that the model identified about two-thirds of the actual survivors. The F1-score of 72.4% represents the balance between precision and recall. The ROC-AUC score of 84.4% indicates that the model has a good ability to distinguish between survivors and non-survivors.

4.2 ROC Curve Analysis

```
from sklearn.metrics import roc_curve, roc_auc_score

y_prob = model.predict_proba(X_test)[:, 1]
fpr, tpr, thresholds = roc_curve(y_test, y_prob)

plt.figure(figsize=(7, 5))
plt.plot(fpr, tpr, label=f"AUC = {roc_auc_score(y_test, y_prob):.3f}")
plt.plot([0, 1], [0, 1], linestyle="--")
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve - Logistic Regression")
plt.legend()
plt.show()
```

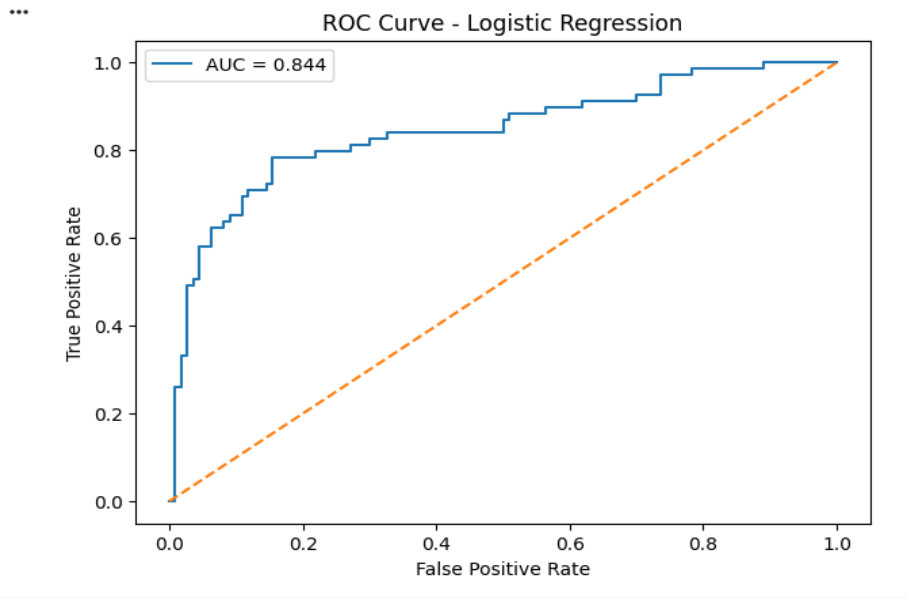


Figure 12. ROC Curve – Logistic Regression

The Receiver Operating Characteristic (ROC) curve is used to evaluate the ability of the Logistic Regression model to distinguish between passengers who survived and those who did not survive. It plots the True Positive Rate against the False Positive Rate at different classification thresholds.

Interpretation: The model achieved an ROC-AUC score of **0.844**, indicating good ability to distinguish between survivors and non-survivors. The ROC curve provides a visual representation of the model's classification performance.

4.3 Feature Coefficient Interpretation

The Logistic Regression model uses feature coefficients to understand the influence of different passenger characteristics on survival predictions. A positive coefficient indicates an increase in the predicted odds of survival, whereas a negative coefficient indicates a decrease in the predicted odds of survival, keeping other features constant.

The model analyzes features such as passenger class, gender, age, family information, ticket fare, and embarkation point to make predictions. The coefficient analysis helps explain how

these features contribute to the model's predictions and provides a better understanding of its decision-making process.

```
# Get feature names after preprocessing
feature_names = model.named_steps["preprocessor"].get_feature_names_out()

# Get coefficients from Logistic Regression
coefficients = model.named_steps["classifier"].coef_[0]

coef_df = pd.DataFrame({
    "Feature": feature_names,
    "Coefficient": coefficients
})

coef_df["Effect"] = np.where(
    coef_df["Coefficient"] > 0,
    "Increases predicted survival odds",
    "Decreases predicted survival odds"
)

display(coef_df.sort_values("Coefficient", ascending=False))
```

	Feature	Coefficient	Effect
5	cat__Sex_female	1.299444	Increases predicted survival odds
8	cat__Embarked_Q	0.304784	Increases predicted survival odds
4	num__Fare	0.100269	Increases predicted survival odds
7	cat__Embarked_C	0.032247	Increases predicted survival odds
3	num__Parch	-0.066424	Decreases predicted survival odds
2	num__SibSp	-0.260607	Decreases predicted survival odds
9	cat__Embarked_S	-0.358305	Decreases predicted survival odds
1	num__Age	-0.502815	Decreases predicted survival odds
0	num__Pclass	-0.921972	Decreases predicted survival odds
6	cat__Sex_male	-1.320718	Decreases predicted survival odds

Fig 13:Feature Coefficient Interpretation

5. Conclusion

The Titanic Survival Prediction project successfully demonstrates the use of Logistic Regression to predict passenger survival based on passenger characteristics. The dataset was analyzed using Exploratory Data Analysis, and missing values and categorical variables were handled during preprocessing. The model achieved an accuracy of 80.4% and an ROC-AUC score of 84.4%, indicating good classification performance on the test dataset. The visualizations and feature coefficient analysis helped identify patterns in the data and understand the influence of different features on survival predictions. Finally, the trained model was deployed using Streamlit, providing a simple web interface for users to enter passenger details and obtain predictions. This project demonstrates how machine learning can be applied to a real-world classification problem.

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1. PROBLEM STATEMENT

The Bank Marketing Prediction case study focuses on predicting whether a customer will subscribe to a term deposit based on customer information and previous marketing campaign details.

The main objective is to develop and compare four classification algorithms: Logistic Regression, Decision Tree, K-Nearest Neighbors (KNN), and Random Forest. The dataset is analyzed, preprocessed, and used to train the models. Their performance is evaluated using accuracy, precision, recall, F1-score, and ROC-AUC to identify a suitable model for predicting customer subscriptions and improving bank marketing campaigns.

2. METHODOLOGY

The methodology followed in this project consists of several steps, beginning with dataset loading and ending with model evaluation.

Step 1: Dataset Loading

The bank marketing dataset is loaded into a Pandas DataFrame using Python. The dataset is examined to understand its dimensions, features, data types, and descriptive statistics.

Descriptive statistics:

	count	unique	top	freq	mean	std	min	25%	50%	75%	max
age	45211.0	NaN	NaN	NaN	40.93621	10.618762	18.0	33.0	39.0	48.0	95.0
job	45211	12	blue-collar	9732	NaN	NaN	NaN	NaN	NaN	NaN	NaN
marital	45211	3	married	27214	NaN	NaN	NaN	NaN	NaN	NaN	NaN
education	45211	4	secondary	23202	NaN	NaN	NaN	NaN	NaN	NaN	NaN
default	45211	2	no	44396	NaN	NaN	NaN	NaN	NaN	NaN	NaN
balance	45211.0	NaN	NaN	NaN	1362.272058	3044.765829	-8019.0	72.0	448.0	1428.0	102127.0
housing	45211	2	yes	25130	NaN	NaN	NaN	NaN	NaN	NaN	NaN
loan	45211	2	no	37967	NaN	NaN	NaN	NaN	NaN	NaN	NaN
contact	45211	3	cellular	29285	NaN	NaN	NaN	NaN	NaN	NaN	NaN
day	45211.0	NaN	NaN	NaN	15.806419	8.322476	1.0	8.0	16.0	21.0	31.0
month	45211	12	may	13766	NaN	NaN	NaN	NaN	NaN	NaN	NaN
duration	45211.0	NaN	NaN	NaN	258.16308	257.527812	0.0	103.0	180.0	319.0	4918.0
campaign	45211.0	NaN	NaN	NaN	2.763841	3.098021	1.0	1.0	2.0	3.0	63.0

Fig 14:Dataset Loading

Step 2: Exploratory Data Analysis

Exploratory Data Analysis (EDA) is performed to understand customer subscription patterns. Graphs are created to examine subscription distribution, subscription rates by job and education, and age differences between subscribers and non-subscribers.

Step 3: Data Preprocessing

Duplicate records are removed, and the target variable is converted into numerical values. Categorical features are converted using One-Hot Encoding, while numerical features are standardized using StandardScaler. The duration attribute is excluded because call duration is not known before contacting a customer

```
print("Missing values in each column:")
print(df.isnull().sum())

print("\nDuplicate rows:", df.duplicated().sum())

print("\nTarget column distribution:")
print(df['y'].value_counts())

print("\nTarget percentages:")
print(df['y'].value_counts(normalize=True).mul(100).round(2))
```

```
Missing values in each column:
age          0
job          0
marital      0
education    0
default      0
balance      0
housing      0
loan         0
contact      0
day          0
month        0
duration     0
campaign     0
pdays        0
previous     0
poutcome     0
y            0
dtype: int64

Duplicate rows: 0

Target column distribution:
y
```

.

Fig 15:Data Preprocessing

Step 4: Model Building

The dataset is divided into 80% training data and 20% testing data. Four classification models—Logistic Regression, Decision Tree, K-Nearest Neighbors, and Random Forest—are trained using the training data.

```
Training features: (36168, 15)
Testing features: (9043, 15)

Training target distribution:
y
0    88.3
1    11.7
Name: proportion, dtype: float64

Testing target distribution:
y
0    88.3
1    11.7
Name: proportion, dtype: float64
```

Fig 16:Model Building

Step 5: Model Evaluation

The trained models are evaluated using accuracy, precision, recall, F1-score, confusion matrices, and ROC-AUC. Performance comparison graphs and ROC curves are used to compare their classification performance.

	Model	Accuracy	Precision	Recall	F1-Score
0	Logistic Regression	0.8933	0.6632	0.1786	0.2815
1	Decision Tree	0.8309	0.2982	0.3289	0.3128
2	KNN	0.8874	0.5529	0.1975	0.2911
3	Random Forest	0.8954	0.6373	0.2457	0.3547

Fig 17:Model Evaluation

3. STATISTICAL FINDINGS

3.1 Target Variable Distribution

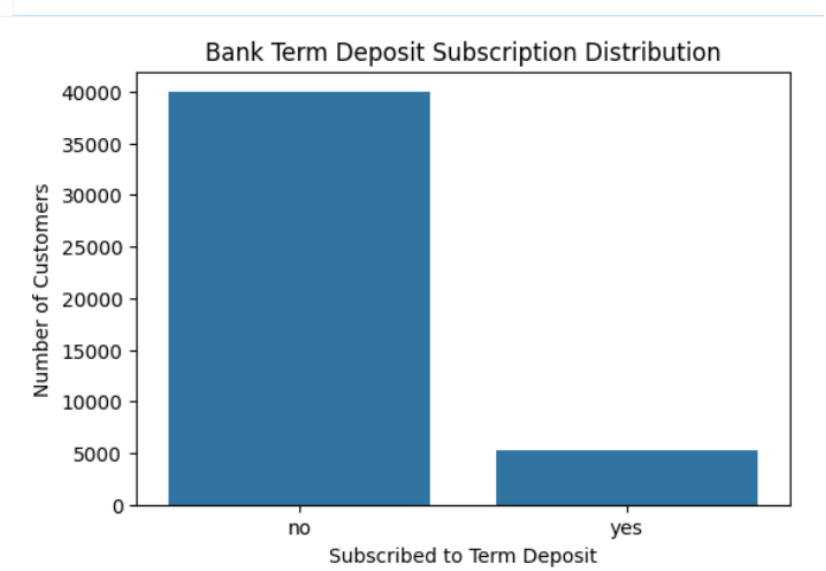


Figure 18: Distribution of Bank Marketing Subscriptions

The target variable represents whether a customer subscribed to a term deposit. The bar chart displays the number of customers who subscribed (yes) and those who did not subscribe (no). The distribution shows that non-subscribers form the majority of the dataset, indicating a class imbalance. Therefore, model evaluation should consider precision, recall, and F1-score along with accuracy.

3.2 Subscription Rate by Job

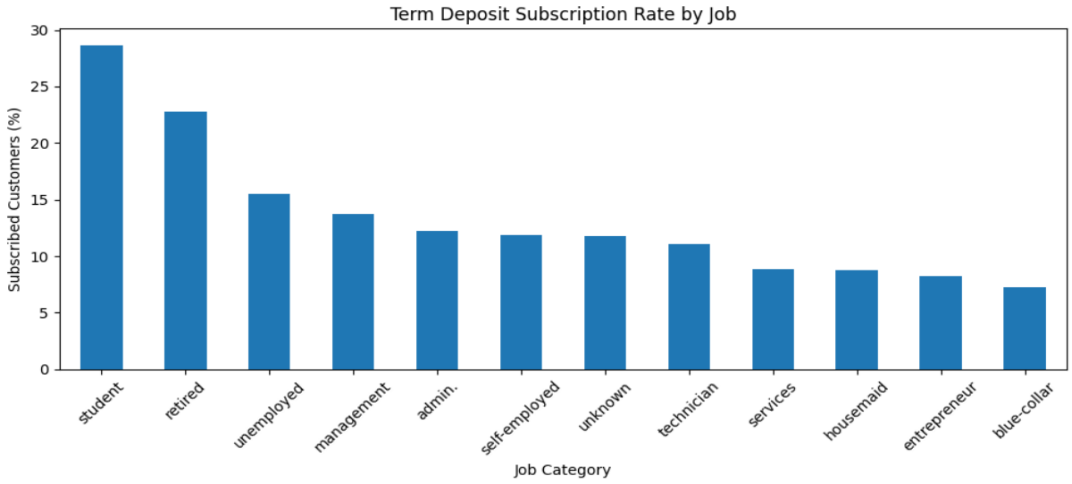


Figure 19: Term Deposit Subscription Rate by Job

The bar chart compares the subscription rates of customers belonging to different job categories. It helps identify which occupational groups show higher or lower interest in term

deposits. This analysis helps the bank understand customer preferences across job categories and plan more effective marketing campaigns.

3.3 Subscription Rate by Education

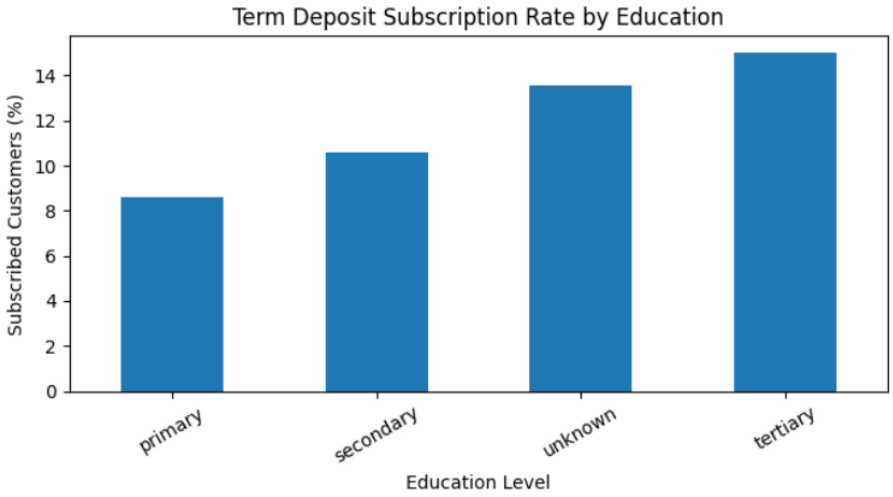


Figure 20: Term Deposit Subscription Rate by Education Level

The bar chart compares the term deposit subscription rates among customers with different education levels. It helps identify differences in subscription behavior across educational groups. This information can help the bank understand customer responses and plan suitable marketing strategies for different education categories.

3.4 Relationship Between Age and Subscription

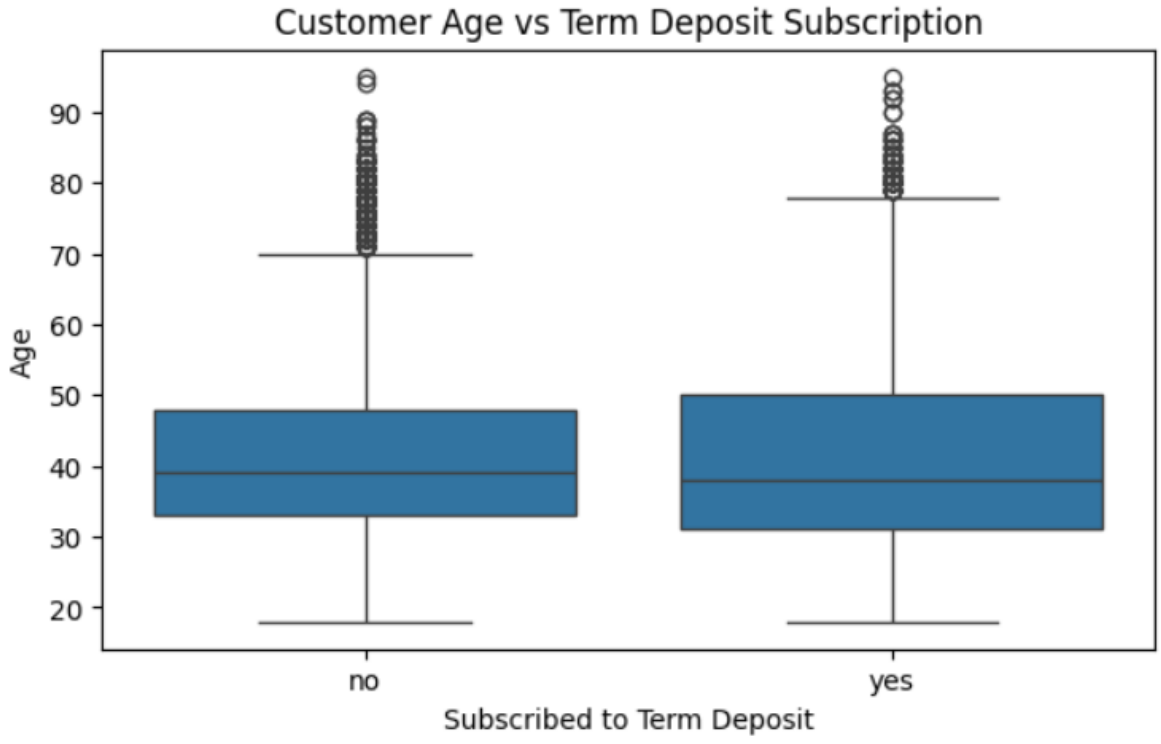


Figure 21: Age Distribution of Customers Based on Term Deposit Subscription

The box plot compares the age distribution of customers who subscribed to a term deposit with those who did not. It helps examine differences in the median age, spread, and distribution of ages between the two groups. This analysis provides insight into customer age patterns and helps the bank understand whether subscription behavior varies across age groups.

4. MODEL EVALUATION

4.1 Performance Comparison of Classification Models

Four classification algorithms—Logistic Regression, Decision Tree, K-Nearest Neighbors (KNN), and Random Forest—were evaluated using accuracy, precision, recall, F1-score, and ROC-AUC. These metrics help compare the models and understand their ability to identify customers who may subscribe to a term deposit

	Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
0	Logistic Regression	0.8933	0.6632	0.1786	0.2815	0.7717
1	Decision Tree	0.8309	0.2982	0.3289	0.3128	0.6132
2	KNN	0.8874	0.5529	0.1975	0.2911	0.7027
3	Random Forest	0.8954	0.6373	0.2457	0.3547	0.7888

Table 1: Performance Comparison of Classification Models

Interpretation: Random Forest achieved the highest accuracy (89.54%), F1-score (0.3547), and ROC-AUC (0.7888) among the four models. Decision Tree achieved the highest recall (32.89%), while Logistic Regression achieved the highest precision (66.32%). Therefore, Random Forest provides the strongest overall balance among these models based on the reported metrics, although all four models have relatively low recall.

4.2 Confusion Matrix Analysis

Confusion matrices were generated for all four classification models to evaluate their prediction performance. They show the number of correct and incorrect predictions for customers who subscribed and those who did not subscribe to a term deposit. These matrices help identify false positives and false negatives and provide a clearer understanding of each model's classification ability.

 Figure 22: Confusion Matrix – Logistic Regression

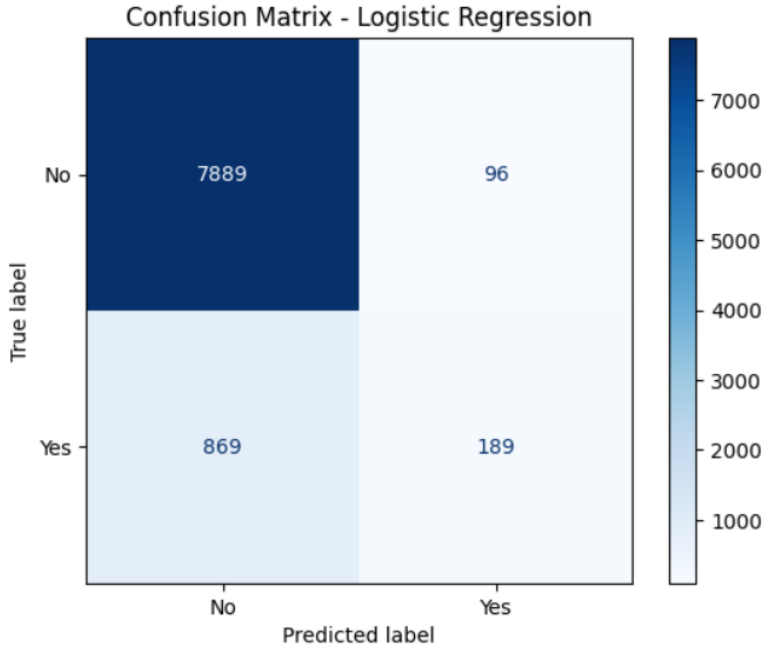


 Figure 23: Confusion Matrix – Decision Tree

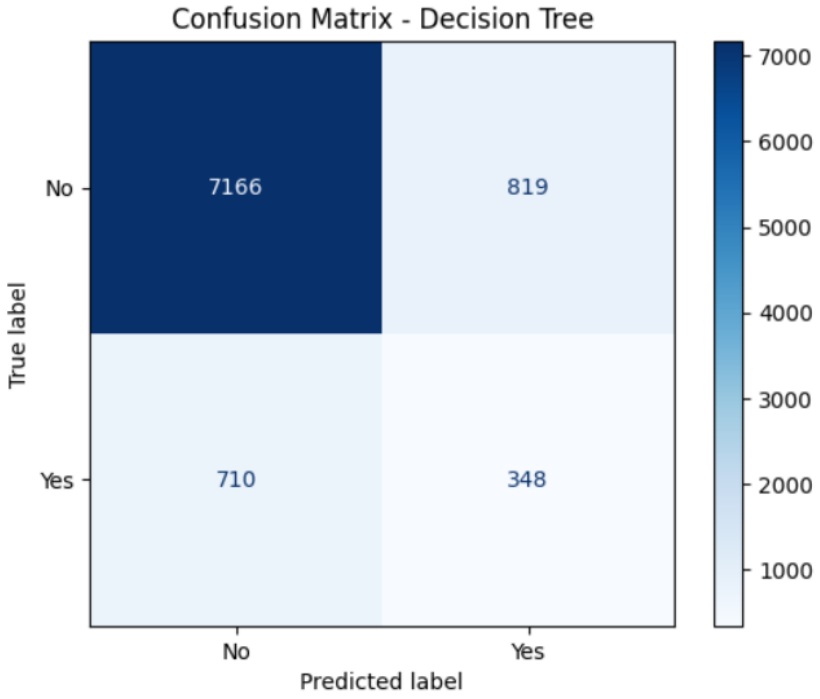


 Figure 24: Confusion Matrix – K-Nearest Neighbors (KNN)

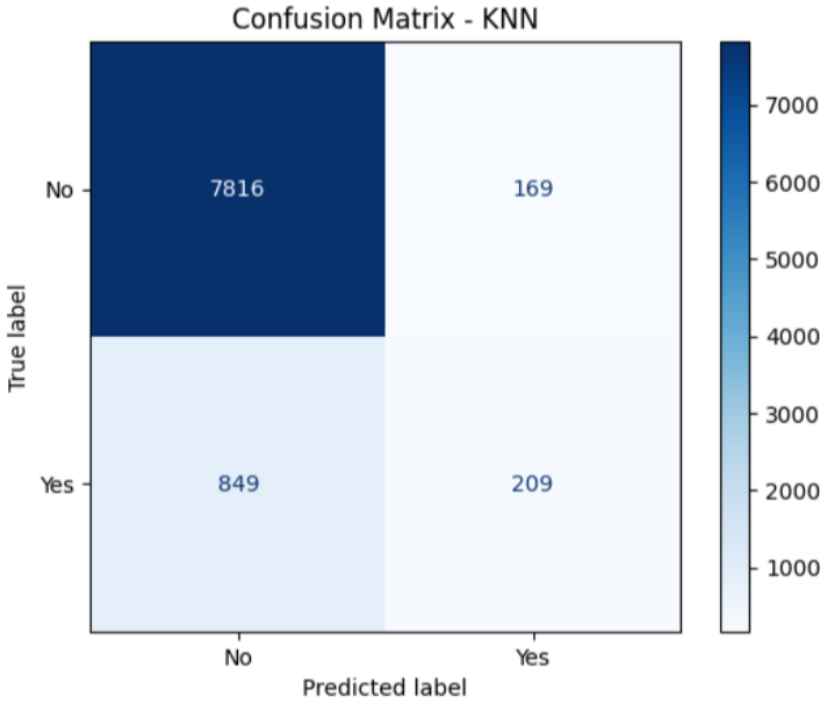
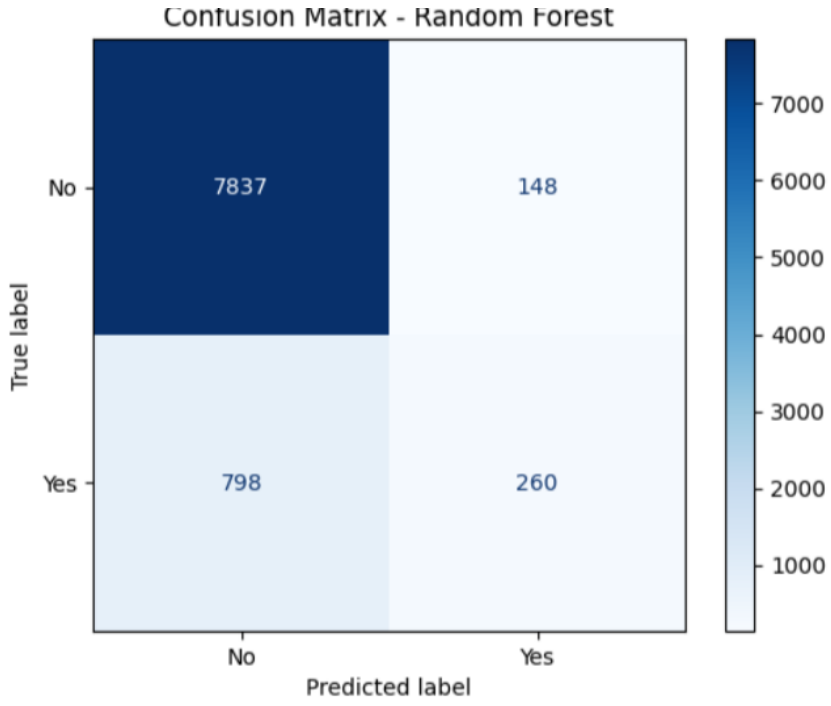


 Figure 25: Confusion Matrix – Random Forest



4.3 ROC Curve Analysis

The Receiver Operating Characteristic (ROC) curve is used to evaluate the ability of classification models to distinguish between customers who subscribed to a term deposit and those who did not. The ROC curve plots the True Positive Rate against the False Positive

Rate at different classification thresholds. The Area Under the Curve (ROC-AUC) summarizes the model's ability to distinguish between the two classes.

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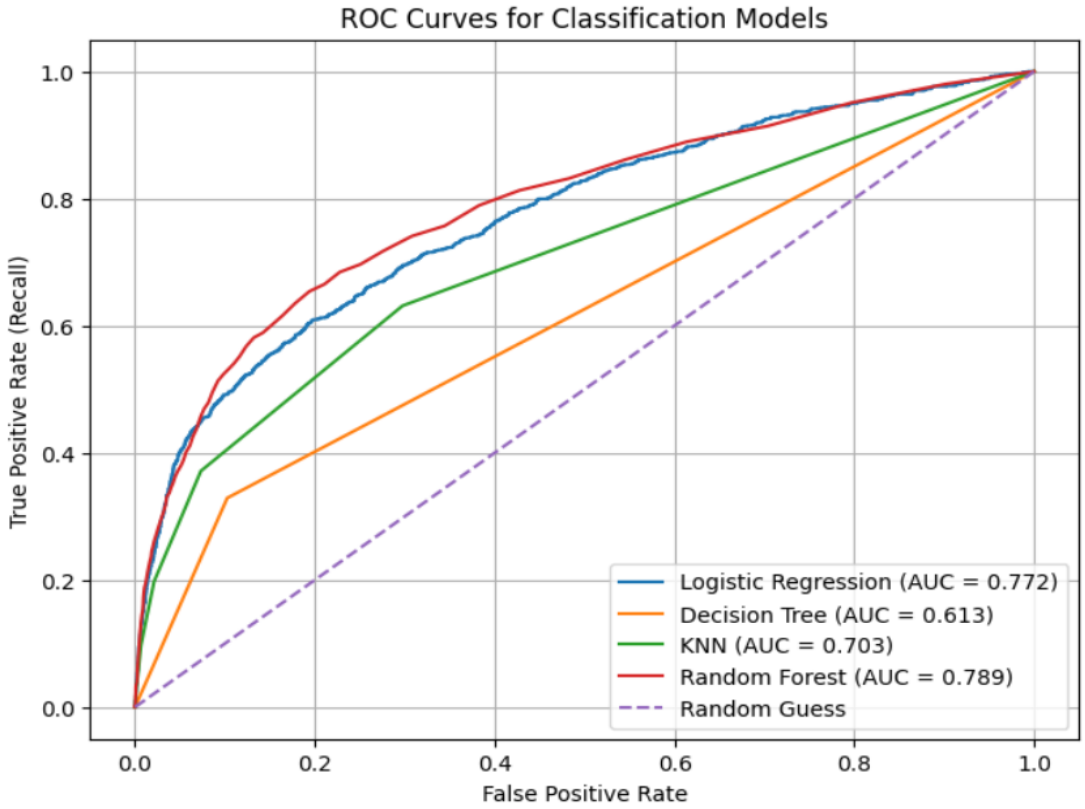


Figure 26: ROC Curve Comparison of Classification Models

Interpretation: Random Forest achieved the highest ROC-AUC of 0.7888, followed by Logistic Regression (0.7717), KNN (0.7027), and Decision Tree (0.6132). These results indicate that Random Forest showed the strongest class-discrimination ability among the four evaluated models.

5. CONCLUSION

This case study compared four classification algorithms for predicting customer subscription to bank term deposits. The models were evaluated using accuracy, precision, recall, F1-score, confusion matrices, and ROC-AUC.

Among the evaluated models, Random Forest achieved the highest accuracy of 89.54%, F1-score of 0.3547, and ROC-AUC of 0.7888. Decision Tree achieved the highest recall of 32.89%, while Logistic Regression achieved the highest precision of 66.32%. These results show that model performance depends on the evaluation metric considered.

The analysis also revealed that most customers in the dataset did not subscribe to a term deposit. Therefore, accuracy alone is not sufficient to evaluate the models. The bank can use these findings to improve customer targeting and make marketing campaigns more effective.

Limitations: The dataset contains an imbalanced target variable, which can affect model performance. In addition, customer behavior may change over time, so the models should be evaluated on updated data before being used in real campaigns.